OBSERVATION OF STIMULATED BRILLOUIN SCATTERING IN LOW-LOSS SILICA FIBRE AT 1·3 μm

Indexing terms: Optical fibres, Scattering

Stimulated Brillouin scattering (SBS) in low-loss single-mode silica fibre is observed at 1·32 μm using a continuous-wave single-frequency Nd:YAG laser. The threshold for SBS is 5 mW and the transmitted power reaches a saturated maximum for launch powers exceeding about 10 mW. A conversion efficiency of 65% is observed. The Stokes frequency shift is 12·7 ± 0·2 GHz.

Introduction: It has been realised for many years that for transmission of narrowband laser radiation in fibres the optical power levels will be limited by nonlinear processes, principally stimulated Brillouin scattering (SBS).[1] SBS was observed in early experiments by Ippen and Stolen[2] but only at input powers exceeding 1 W. For this reason SBS has been widely disregarded as a significant loss mechanism for optical communication systems which generally operate at much lower power levels. More recent experiments, however, have demonstrated that in long length fibres of lower loss the power levels necessary to exceed the SBS threshold are reduced.[3] However, there have been no reports of SBS in the long-wavelength region (1·2–1·6 μm), in which silica fibres have lowest loss and future coherent optical transmission systems using narrowband lasers are most likely to be used.

We have detected SBS in 13 km and 32 km lengths of silica fibre at 1·32 μm. This is the first reported observation of SBS in a low-loss single-mode fibre (i.e. at a wavelength greater than the cutoff) and thus the possibility of coupling to higher-order modes has been excluded. The input power required to exceed the threshold for nonlinear scattering was found to be around 5 mW. To our knowledge this is the lowest power level at which any nonlinear optical scattering process has been observed.

Theory: According to the small-signal steady-state theory of stimulated scattering[1,4] the maximum laser power P_L which can be launched into the fibre before SBS becomes detectable is determined by

$$GL_e \simeq 21 \tag{1}$$

where G is the SBS gain factor

$$G = \frac{2\pi n^7 p_{12}^2 K}{c\lambda^2 \rho_0 v_a \Delta\nu_B}\left(\frac{P_L}{A}\right)\ \mathrm{m}^{-1} \tag{2}$$

and where n is the refractive index, ρ_0 is the material density, v_a is the acoustic velocity and p_{12} is the longitudinal elasto-optic coefficient. A is an effective cross-sectional area of the guided mode such that the peak intensity is given by P_L/A. An assumption here is that the laser linewidth is small compared to $\Delta\nu_B$, which is the linewidth for spontaneous Brillouin scattering (Hz, FWHM) at ambient temperature. The factor K is unity for a fibre which maintains the optical polarisation, and is 1/2 otherwise.[5] The effective interaction length is given by

$$L_e = \alpha^{-1}(1 - \exp[-\alpha L]) \tag{3}$$

where α is the absorption coefficient (m^{-1}) and L is the fibre length. For long fibre lengths used in communications it is usual for $L \gg \alpha^{-1}$, and hence $L_e \simeq \alpha^{-1}$. Low-loss fibres have longer interaction lengths and thus have lower SBS thresholds.

We insert the following bulk parameters for fused silica[6,7]: $n = 1{\cdot}451$, $\rho_0 = 2{\cdot}21 \times 10^3\ \mathrm{kg\,m^{-3}}$, $v_a = 5{\cdot}96 \times 10^3\ \mathrm{m\,s^{-1}}$, $p_{12} = 0{\cdot}286$. The spontaneous linewidth $\Delta\nu_B$ is 38·4 MHz at the wavelength $\lambda = 1{\cdot}0$ μm and varies as λ^{-2};[8] therefore we assume $\Delta\nu_B = 22$ MHz at 1·32 μm. Finally we insert the following values appropriate to our 13·6 km test fibre: $\alpha = 9{\cdot}5 \times 10^{-5}\ \mathrm{m}^{-1}$ (0·41 dB/km loss), $L_e = 7{\cdot}6$ km, $A = 4{\cdot}7 \times 10^{-11}\ \mathrm{m}^2$ and $K = 1/2$. Thus we predict the threshold for SBS in this test fibre at 1·32 μm to be $P_L \simeq 5{\cdot}6$ mW.

Experiment: The experiments were carried out using a continuous-wave single-frequency Nd:YAG laser operating on the 1·319 μm transition.[9] This laser produces an output power of around 100 mW in a single longitudinal mode and diffraction-limited TEM_{00} transverse mode. The laser linewidth was measured using a scanning confocal Fabry–Perot interferometer of 300 MHz free spectral range, and found to be less than 1·6 MHz which is the instrumental resolution. This is more than an order of ten narrower than the spontaneous Brillouin linewidth $\Delta\nu_B$. As shown in Fig. 1 the laser output was attenuated using a circular variable density filter, and focused into the test fibre using a microscope objective.

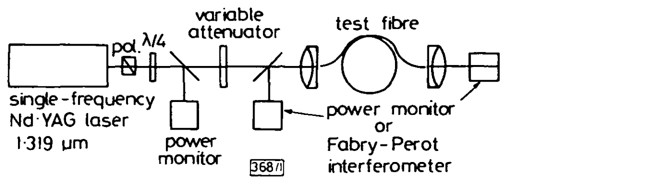

Fig. 1 *Experimental arrangement*

The optical power emitted from both the near and distant ends of the fibre was monitored using calibrated Ge photodiodes. A scanning confocal Fabry–Perot interferometer of 7·5 GHz free spectral range was used to record the frequency spectrum of the emitted light. At the conclusion of the experiments on each fibre, the fibre was cut back to within a few metres of the launch objective to measure the power propagating in the guided mode.

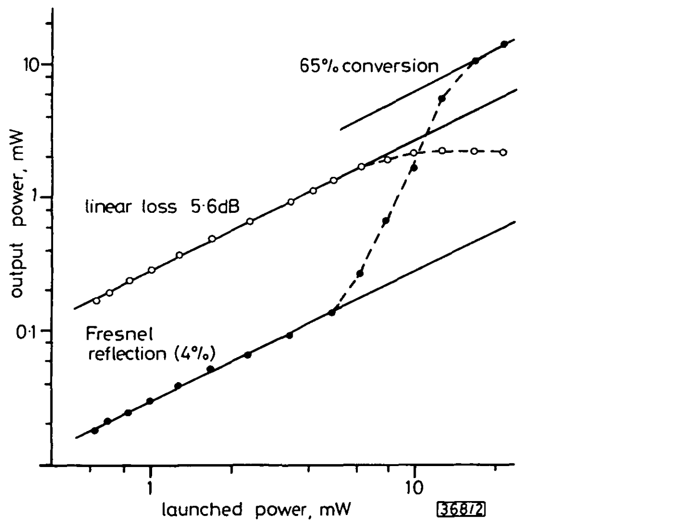

Fig. 2 *Power emitted from distant (forward) and near (backward) ends of 13·6 km single-mode fibre as a function of launched power*

○ forward ● backward

The first experiments were carried out using a 13·6 km length of GeO_2-doped silica fibre having a core diameter of 9 μm, core-cladding index difference of 0·3%, cutoff wavelength of 1·21 μm, and loss at 1·32 μm of 0·41 dB/km. Using the measured refractive index profile a computer solution of the guided mode distribution gives $A = 4{\cdot}7 \times 10^{-11}\ \mathrm{m}^2$ at 1·32 μm. Fig. 2 shows the output power from each end of the fibre as a function of the launched power. At low input power the output power monitored in the backward direction was due only to the Fresnel reflection from the cleaved end face of the fibre. However, at input powers exceeding 5 mW, the output power in the backward direction was seen to increase rapidly in a nonlinear fashion, and the efficiency of conversion to the backscattered wave reached 65%. At low input powers the power emitted from the distant end of the fibre was related linearly to the input power, determined by the linear loss of 5·6 dB. However, at input powers exceeding 6 mW, the output power became nonlinear. At input powers greater than 10 mW the output power in the forward direction reached a saturated maximum of about 2 mW.

Fig. 3 shows the Fabry–Perot spectrum of the backscattered light. A small amount of laser light was deliberately coupled into the interferometer to provide a calibration marker. The spectral component designated 'Stokes' was present only when the power launched into the fibre exceeded the threshold value of 5 mW. If, as shown in Fig. 3, it is assumed that the laser and backscatter signals are separated by two interferometric orders (free spectral range 7·5 GHz), then the Stokes shift is 12·7 ± 0·2 GHz. This is in good agreement with the predicted value of 13·1 GHz calculated from the expression $2v_a n/\lambda$, where the symbols are as defined previously, taking the value of $5{\cdot}96 \times 10^3$ m s^{-1} for the acoustic velocity in fused silica.[6] The displayed linewidths are limited by the resolution of the measurement.

The frequency spectrum of light emitted from the distant end of the fibre comprised an intense component at the laser frequency and a weaker component at the Stokes frequency, the latter probably being due to reflection from the laser output mirror. Surprisingly no anti-Stokes or higher-order

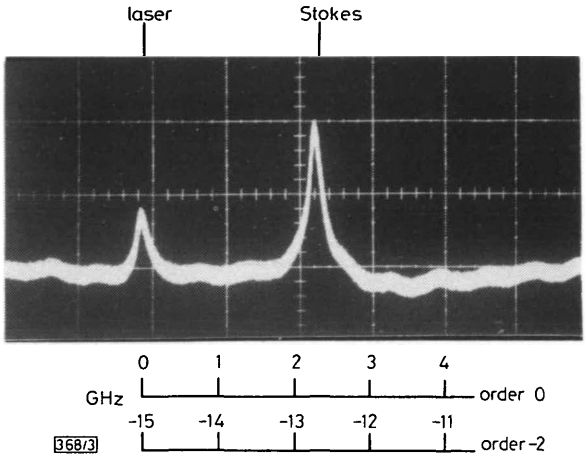

Fig. 3 *Fabry–Perot spectrum of backscattered light; free spectral range 7·5 GHz*

Stokes emissions were observed in this experiment despite the pressure of feedback from the laser optics.[10]

The linear polariser and quarterwave plate shown in Fig. 1 were intended to provide optical isolation between the laser and fibre. However, under conditions of strong SBS, this arrangement proved ineffective in isolating the laser from the backscattered signal, due to polarisation scrambling in the fibre. Nevertheless, the laser continued to operate in a stable single longitudinal mode under all conditions, probably because the frequency of the backscattered light was shifted sufficiently from the peak of the Nd : YAG gain curve.

Experiments were also carried out using a 31·6 km length of cabled single-mode fibre having a total linear loss of 17·4 dB at 1·32 μm. The experimental results were similar to those found in the 13·6 km fibre, SBS being observed with launched powers greater than 6 mW. Despite the longer physical length, the effective interaction length L_e (eqn. 3) of 7·7 km for the 31·6 km cabled fibre is almost identical to that of the 13·6 km fibre. The other fibre parameters are similar, and thus it is to be expected from eqn. 1 that the SBS threshold power will be closely similar for the two fibres.

Conclusions: We have shown that SBS can occur in low-loss fibres at power levels of only a few milliwatts and that the efficiency of conversion to the backscattered light can reach 65%. The observed threshold powers are in good agreement with those predicted from eqn. 1 taking the bulk acoustic and elasto-optic parameters of fused silica. Evidently the doping levels used in low-loss fibres do not significantly influence the SBS threshold even when the laser linewidth is much narrower than the spontaneous Brillouin scattering linewidth.[5] At 1·32 μm the Stokes frequency shift is 12·7 ± 0·2 GHz in fairly good agreement with theory. Comparative measurements using a 13·6 km and 31·6 km fibres are in good agreement with the expected dependence on fibre length.

It has been proposed recently[11] that fibres manufactured from materials having an infra-red absorption edge at a wavelength longer than that of silica would have very low intrinsic loss, perhaps as small as 0·001 dB/km. We predict, however, that, if sufficiently long lengths of such fibres could be manufactured, the threshold powers for SBS could be as low as a few microwatts.

It has become clear that SBS could present a severe practical limitation for future coherent optical data transmission systems. We are currently studying techniques whereby this limitation could be overcome.[12]

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REMARKS ON SYSTEM SIMPLIFICATION UNDER CONSIDERATION OF TIME RESPONSES

Indexing terms: Modelling, Linear systems, Model reduction

The letter highlights the significance of matching the terms of expansions around several points including $s = 0$ and $s = \infty$ in realising improved overall time-response approximation.

It is well known that, for a certain class of inputs,[1] the matching of terms of expansions around the points $s = 0$ (time moments) and $s = \infty$ (Markov parameters) of the original transfer function and its reduced-order equivalent leads to steady-state and initial-time-response matching. It is intuitively felt that if the matching of expansions around some additional (finite) points is also taken into account, this may result in good overall time response approximation. The present letter is intended to illustrate this.

Consider a third-order system given by[2]

$$G(s) = \frac{2 + 6s + 8s^2}{2 + 5s + 4s^2 + s^3} \quad (1)$$